MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5750

The complete requirements for procuring the electron tube described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Miniature, pentagrid converter

Outline --- 5-2 EIA

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	1	2	3	4	5	6	7
Element	G1	k,g5	h	h	p	g2,g4	g3

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2 and 4	Ec3	Ehk
Unit:	V	Vdc	Vdc	Vdc	Vdc	v
Maximum:	6.9	330	---	110	0, -55	100
Minimum:	5.7	---	---	---	---	---
TEST CONDITIONS:	6.3	250	0	100	-1.5	---

ABSOLUTE RATINGS:

Parameter:	Ik	Ic1	Pp	Pg2 and 4	TE	Alt
Unit:	mAdc	mAdc	W	W	°C	ft
Maximum:	15.5	1.0	1.2	1.2	165	(See note 11)
Minimum:	----	---	---	---	---	
Test Conditions:	----	---	---	---	---	---

GENERAL

Qualification - Required

Reliable tube

Ⓖ DENOTES CHANGES

FSC 5960

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TABLE I Quality Conformance Inspection.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	MIN	MAX	UNIT
	Quality conformance inspection, part 1							
1256	Plate current (1)	See note 4	0.4	II	Ib	1.5	3.5	mAdc
1256	Plate current (2)	Eb = 100 Vdc See note 5 Ec1 = -14 Vdc; Ec3 = 0	0.4	II	Ib	---	50	μAdc
1260	Total grid 3 current	Ec3 = -2 Vdc See notes 1 & 4	0.4	II	Ic3	0	-1.0	μAdc
1301	Heater current		0.4	II	If	275	325	mA
1306	Transconductance (1) (oscillator)	Eb = 100 Vdc; Ec3 = 0 See note 5	0.4	II	Sm	6000	9600	μmhos
1326	Transconductance (1) (conversion)	See note 4	0.4	II	Sc	300	700	μmhos
1336	Heater-cathode leakage		0.4	II	Ihk	---	10	μAdc
1201.5	Shorts and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1105	Permanence of marking		---	---	---	---	---	---
1211	Insulation of electrodes		2.5	S3	R	100	---	Meg
1031	Low-frequency vibration	Ec1=Ec3= -3Vdc; Rp = 10,000 ohms	6.5	Code H	Ep	---	300	mVac
----	Blocking	Ec1 = -3 Vdc See note 3; Rg3 = 0 (except for meter resistance)	2.5	S3	---	---	---	---
1256	Screen-grid current	See note 4	2.5	I	Ic2 and Ic4	---	10.6	mAdc
1256	Cathode current (oscillator)	Eb = 100 Vdc; Ec 3 = 0 See note 5	6.5	S3	Ik	16	33	mAdc
1306	Transconductance (2) (oscillator)	Ef = 5.7 Vdc; Eb = 100 Vdc; Ec3 = 0 See note 5	2.5	I	$\frac{\Delta Sm}{Ef}$	---	15	%
1316	Amplification factor (oscillator)	Eb = 100 Vdc; Ec3 = 0 See note 5	6.5	S3	Mμ	18.5	26.5	---

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TABLE 1 Quality Conformance Inspection (Continued).

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMIT MIN	LIMIT MAX	UNIT
	Quality conformance inspection, part 2 Continued							
1325	Transconductance (2) (conversion)	Ec3 = -30 Vdc See note 4	2.5	I	Sc	1	50	μmhos
1326	Transconductance (3) (conversion)	Ec3 = -6 Vdc See note 4	2.5	S3	Sc	60	300	μmhos
1331	Direct-interelectrode capacitance	No shield	1.5	Code E	Cg3-p	---	0.30	pf
					Cg3-all	5.7	8.1	pf
					Cp-all	5.1	10.1	pf
1121	Base strain		---	---	---	--	---	---
2126	Glass strain		2.5	I	---	---	---	---
1041	Shock	G = 450; Ec1 = Ec3 = -3 Vdc; Ehk = 100 Vdc; Rg1 = Rg3 = 0.1 meg; See note 10	---	---	---	---	---	---
1031	Vibration-fatigue test	G = 2.5; fixed frequency; F = 25 min, 60 max	6.5	See note 8	---	---	---	---
----	Post-shock and vibration-fatigue test end points	Conversion transconductance (1)	---	---	Sc	250	---	μmhos
		Low-frequency vibration	---	---	Ep	---	450	mVac
		Heater-cathode leakage	---	---	Ihk	---	30	μAdc
		Total grid 3 current	---	---	Ic3	0	-2.0	μAdc
	Quality conformance inspection, part 3							
1506	Heater-cycling life test	Ehk = 135 Vdc; Ef = 7.5V; Eb = Ec1 = Ec2 and 4 = Ec3 = 0; 1 min "on" 4 min "off" See note 1	---	---	---	---	---	---
	Heater-cycling life test end point:							
1336	Heater-cathode leakage		---	---	Ihk	---	20	μAdc
1516	Stability life test	Ec1 = -16 Vdc; Ehk = +135 Vdc; Rg3 = 0.05 meg; TA = room See note 7	---	---	---	---	---	---
	Stability life-test end point:							
1326	Change in conversion transconductance (1) of individual tubes		---	---	$\frac{\Delta Sc}{t}$	---	10	%

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TABLE I Quality Conformance Inspection (Continuec).

METHOD	REQUIREMENT OF TEST	CONDITIONS	Sym	LIMITS Min	LIMITS Max	Unit
	Quality conformance inspection, part 3 continued					
	Survival-rate life test end points:					
1201	Shorts and discontinuity detection		---	---	---	---
1326	Conversion transconductance (1)		Sc	230	---	µmhos
1501	Intermittent life test	Stability life-test conditions: TE = +1[illegible]°C min See notes 2 and 7	---	---	---	---
	Intermittent life test end points: (500 hours)	See note [illegible]				
	Inoperatives		---	---	---	---
12[illegible]	Total grid 3 current		Ic3	0	-1.0	µAdc
1301	Heater current		If	275	325	mA
1326	Conversion transconductance (1)		Sc	2[illegible]0	700	µmhos
1306	Transconductance (1) oscillator		Sm	4[illegible]00	---	µmhos
1336	Heater-cathode leakage		Ihk	---	20	µAdc
1211	Insulation of electrodes	Eb(g-all)= -100Vdc Eb(g3-all)=-100Vdc Eb(p-all)= -300Vdc	R	50	---	Meg
1326	Conversion transconductance (1) average change		Avg ΔSc/t	---	17	%
	Total defectives		---	---	---	---
	Intermittent life-test end points: (1,000 hours)	See note [illegible]				
----	Inoperatives		---	---	---	---
1266	Total grid 3 current		Ic3	0	-1.0	µAdc
1301	Heater current		If	27[illegible]	32[illegible]	mA
1326	Conversion transconductance (1)		Sc	230	700	µmhos
1306	Transconductance (1) oscillator		Sm	4[illegible]00	---	µmhos
1336	Heater-cathode leakage		Ihk	---	20	µAdc
	Total defectives		---	---	---	---

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NOTES:

1. This test to be performed at the conclusion of the holding period.
2. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at minimum specified temperature at any position in the life-test rack.
3. Ic3 shall not become negative when Ec3 is varied from 0 to +30 Vdc.
4. Insert 20,000 ohms from grid 1 to ground. Apply sufficient ac signal to grid 1 to produce 0.3 mAdc average grid current. Other equipment which correlates with method 1326 may be used.
5. Nonoscillating condition. Tie grids 2 and 4 to plate; tie grid 3 to cathode.
6. This test shall be made on a lot-by-lot basis.
7. Apply 16.5 Vac, 60 cycle, signal to grid 1 through a 250-Ohm isolating resistor.
8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
9. If a tube is defective for more than one characteristic, the characteristic appearing first in the life-test end-point column shall constitute the failure.
10. Grid resistors of 0.1 megohm shall be added; however, these resistors shall not be used when a thyratron-type short indicator is employed.
11. See 6.1.2.3 for altitude (low pressure) rating.

Custodians:

Army - ER
Navy - EC
Air Force - 85

Review activities:

Army - ER
Navy - EC
Air Force - 11, 17

Preparing activity:

Air Force - 85

(Project 5960-3307)

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